INNOVATION LAB

Beyond Extraction:

Reconciling industry data with stateful AI

Member of innovationXlab

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Date
19 June 2026

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Sponsored by:
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Introduction to:

Staple and Project FIBULA

Staple AI is a Singapore-based AI document processing company focused on helping enterprises automate document-heavy workflows.

- **Automate document lifecycle**
- **Extracted data is traceable, auditable, and defensible**
- **98+% Document level accuracy**

Fibula is a chat-first, node-based automation workspace that turns natural-language workflow intent into inspectable automation logic developed by Staple

- **Allows non-technical users to describe requirements in plain language**
- **Uses an AI agent to generate node-based workflows**
- **Turns natural-language into visible workflow logic**

AI extraction is not enough for industrial reconciliation

When variances occur, users need more than extracted fields. They need linked evidence, clear validation logic, and actionable next steps. Without this, non-technical operators must manually trace issues across documents, workflows, and exception messages.

- Stateless pipelines break at variance points
- Users need evidence, not just extracted data
- Non-technical users depend on developer support
- Low visibility reduces trust

Can Fibula help non-technical users create, inspect, test, and publish stateful reconciliation workflows without assistance from researchers or developers?

Project scope

This project evaluated how Fibula can support self-service industrial reconciliation workflows for non-technical users. Our work focused on understanding the reconciliation logic, translating it into data extraction and workflow processes, and using iterative user research to identify how the platform could be improved.

01.

Research on reconciliation logic

The team analyzed how AP and BOM reconciliation works, including the key documents, matching fields, validation checks, and review outcomes. This logic was presented to Staple AI and used to support the extraction and reconciliation workflow design.

02.

Iterative User Research

The team tested Fibula with participants using task-based sessions, think-aloud observations, and a no-help rule. Each round helped us understand where users struggled when creating, testing, and interpreting workflows.

03.

Snapshot & Improvement Proposal

The team organised user observations into interview snapshots, identified recurring friction patterns, and proposed improvements

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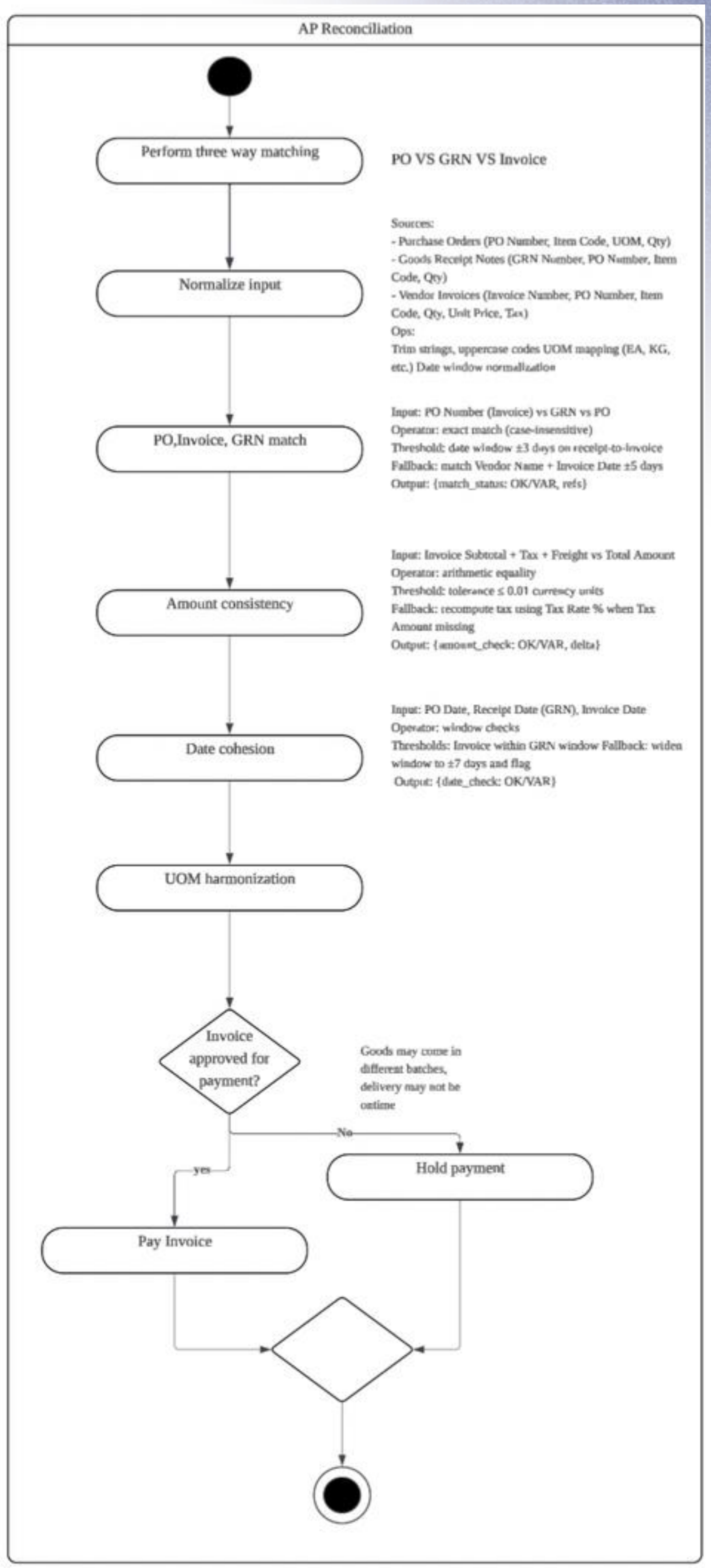
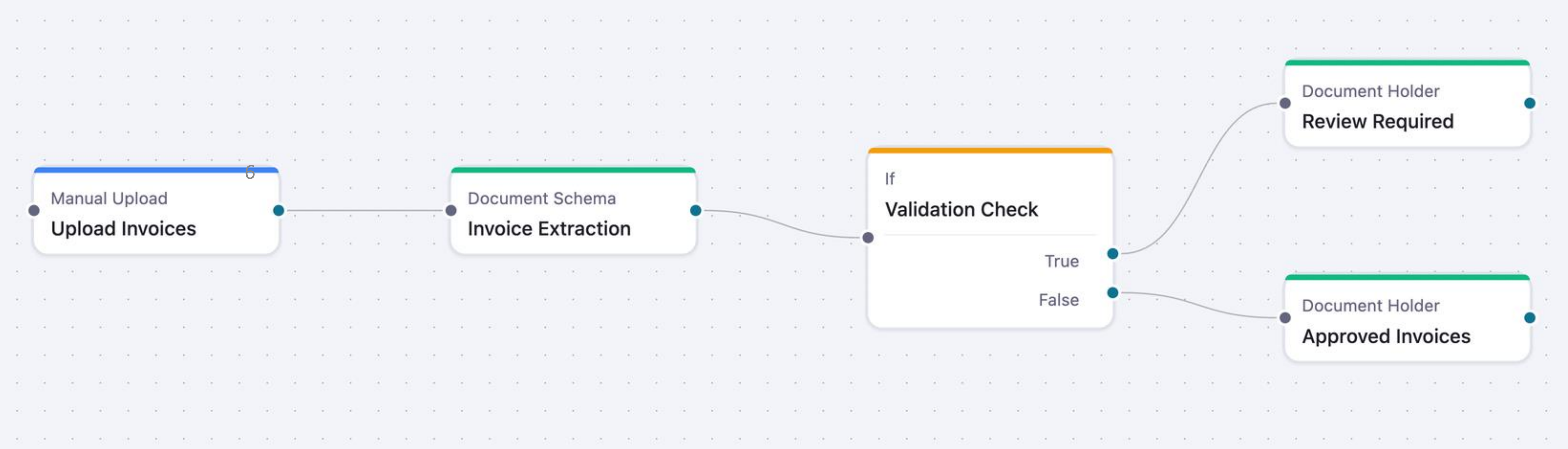


AP Reconciliation
Perform three way matching
PO VS GRN VS Invoice
Normalize input
Sources:
- Purchase Orders (PO Number, Item Code, UOM, Qty)
- Goods Receipt Notes (GRN Number, PO Number, Item Code, Qty)
- Vendor Invoices (Invoice Number, PO Number, Item Code, Qty, Unit Price, Tax)
Ops:
Trim strings, uppercase codes UOM mapping (EA, KG, etc.) Date window normalization
PO,Invoice, GRN match
Input: PO Number (Invoice) vs GRN vs PO
Operator: exact match (case-insensitive)
Threshold: date window ±3 days on receipt-to-invoice
Fallback: match Vendor Name + Invoice Date ±5 days
Output: {match_status: OK/VAR, refs}
Amount consistency
Input: Invoice Subtotal + Tax + Freight vs Total Amount
Operator: arithmetic equality
Threshold: tolerance ≤ 0.01 currency units
Fallback: recompute tax using Tax Rate % when Tax Amount missing
Output: {amount_check: OK/VAR, delta}
Date cohesion
Input: PO Date, Receipt Date (GRN), Invoice Date
Operator: window checks
Thresholds: Invoice within GRN window Fallback: widen window to ±7 days and flag
Output: {date_check: OK/VAR}
UOM harmonization
Invoice approved for payment?
Goods may come in different batches, delivery may not be ontime
No
yes
Hold payment
Pay Invoice

To:



Manual Upload
Upload Invoices
Document Schema
Invoice Extraction
If
Validation Check
True
False
Document Holder
Review Required
Document Holder
Approved Invoices

BOM
Bill of Material

AP
Account payable

Other Reconciliation

Purpose: Check whether what was **ordered**, **received**, and **invoiced** are consistent

Documents included:

- Purchase order
- Good receive notes
- Vendor invoice

Key fields included:
Primary: PO Number
Supporting: Item Code, Quantity, UOM, Unit Price, Currency, Vendor Name, Document Date

Output:
Approved, review required, rejected, or payment held.

BOM
Bill of Material

Purpose: Check whether actual material usage matches expected production consumption.

Documents included:

- BOM
- Work order
- Good issue record
- Rework log

Key fields included:

Primary: Work Order ID

Supporting: BOM Quantity per Unit, Actual Output, GIR Quantity, Scrap/Rework Quantity, Scrap Allowance

Output:

Variance identified, review required, supplier chargeback, or internal improvement action.

Purpose: Purpose: reveal what users expected, misunderstood, or trusted

Participants were asked to verbalize what they were looking for, what they expected to happen, and how they interpreted the workflow.

"I'm currently looking for my workflow…"
"I don't understand why there's multiple theme"
"I'm not sure if the workflow is ending here"

Guiding questions

No-help

Think alou

Purpose: Test whether Fibula could work as a self-service platform.

Researchers did not help when users chose the wrong path or misunderstood. If a user could not progress after 3 attempts, the issue was recorded as a usability failure before a minimal nudge

No-help failure = user needs researcher help to continue, interpret a node, recover from an error, or understand the next action.

Snap shot

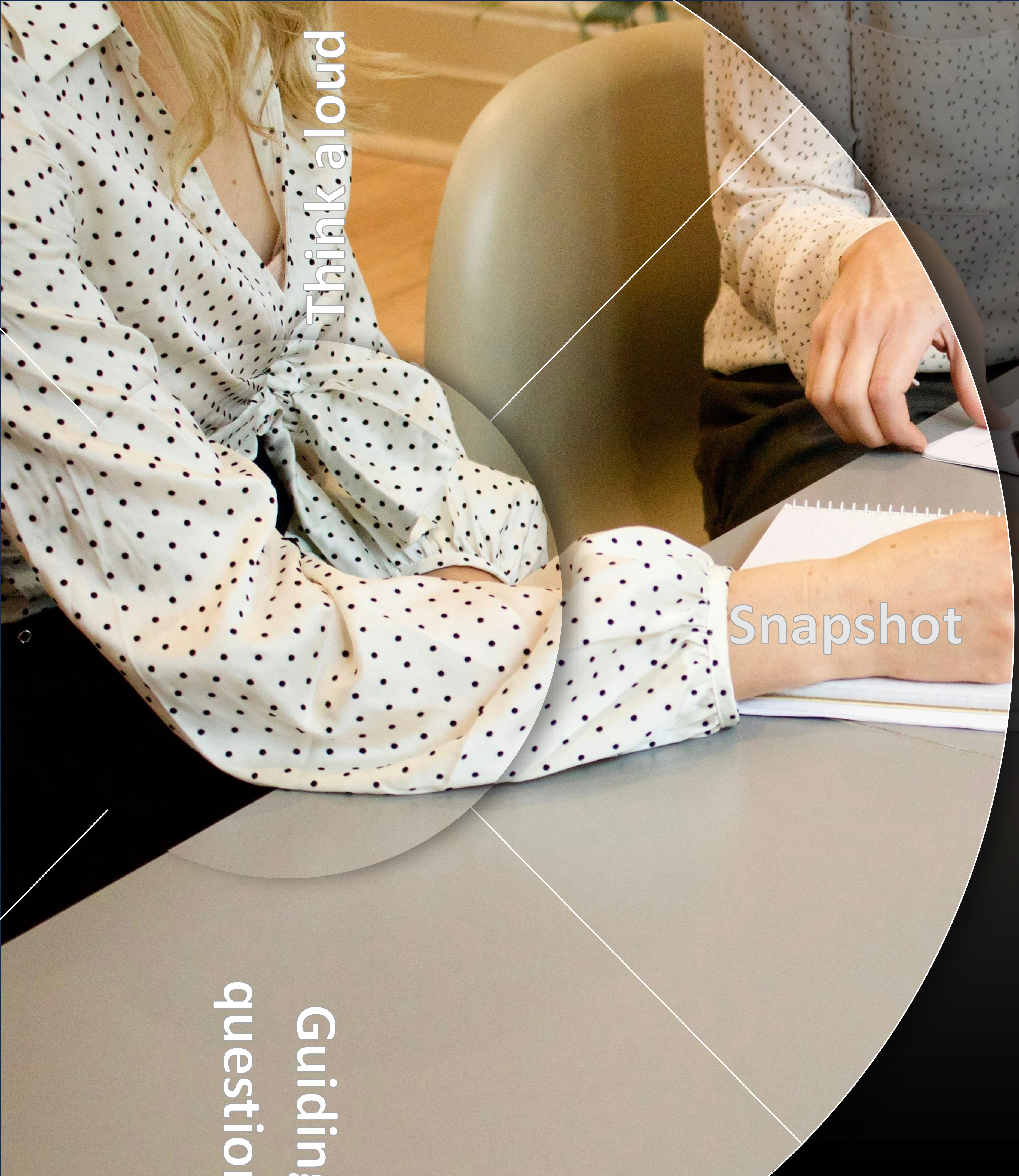
Guiding questions

No-help

Purpose: Help user to think, interpret and understand what the user was thinking without giving the answer directly

the guiding question help the researcher to understand what the user was expecting more in depth and when users were fully blocked, researchers used guiding questions rather than direct instructions,

“why didn’t you try this button?”
“what did you expect this button to do?”
“Were you confused by this information?”



Purpose: Turn raw observations into structured evidence

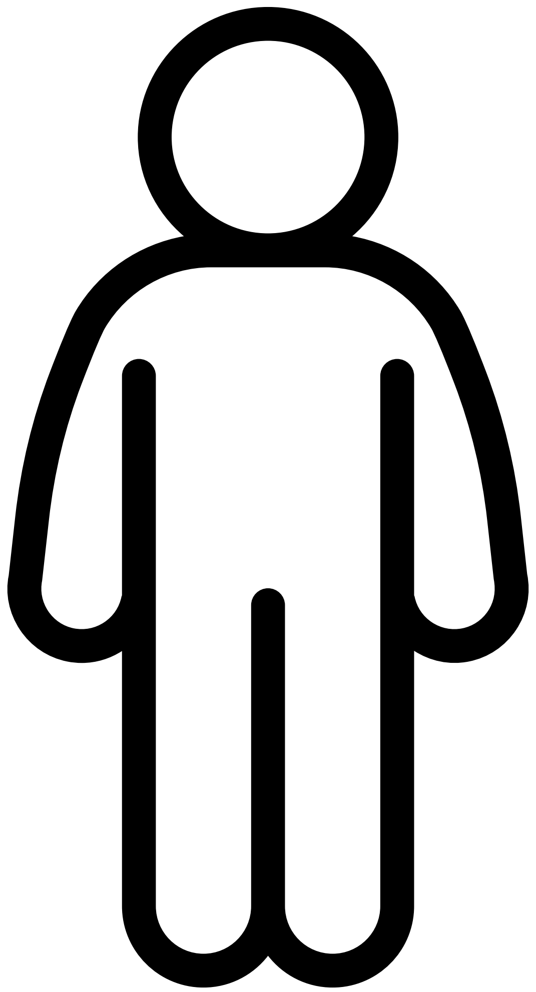
Snapshot include:

- **Quick Facts**
- **Insights**
- **Opportunities**

More organized, without confusing and forgetting how each session went and provide a more structure report for Staple

User research

To evaluate whether Fibula could support non-technical users, we conducted an interview for our user research. The goal was to observe whether users could understand the configuration workflows without researcher or developer support.



Our participants

15 participants coming mainly from non-technical/semi-technical user

Interview format

semi-structured task-based interview, user are encouraging to explore

Reseacher’s task

Record, observe, ask guiding questions

Insights and finding

Each interview session will be organized into an interview snapshot

User research iterative

Participants create or modify workflows using the AI agent, builder, templates, and document upload features.

Researchers convert observations into interview snapshots, identifying quick facts, user struggles, recurring friction points, and opportunities for improvement.

ON BOARDING

Introduce the study purpose, explain that Fibula is being tested rather than the user, explain tasks

WORKFLOW BUILDING

WORKFLOW VALIDATING

Participants test the generated workflow and interpret whether it works, whether the output is clear, and their feeling/suggestions.

INSIGHT & OPPORTUNITIES

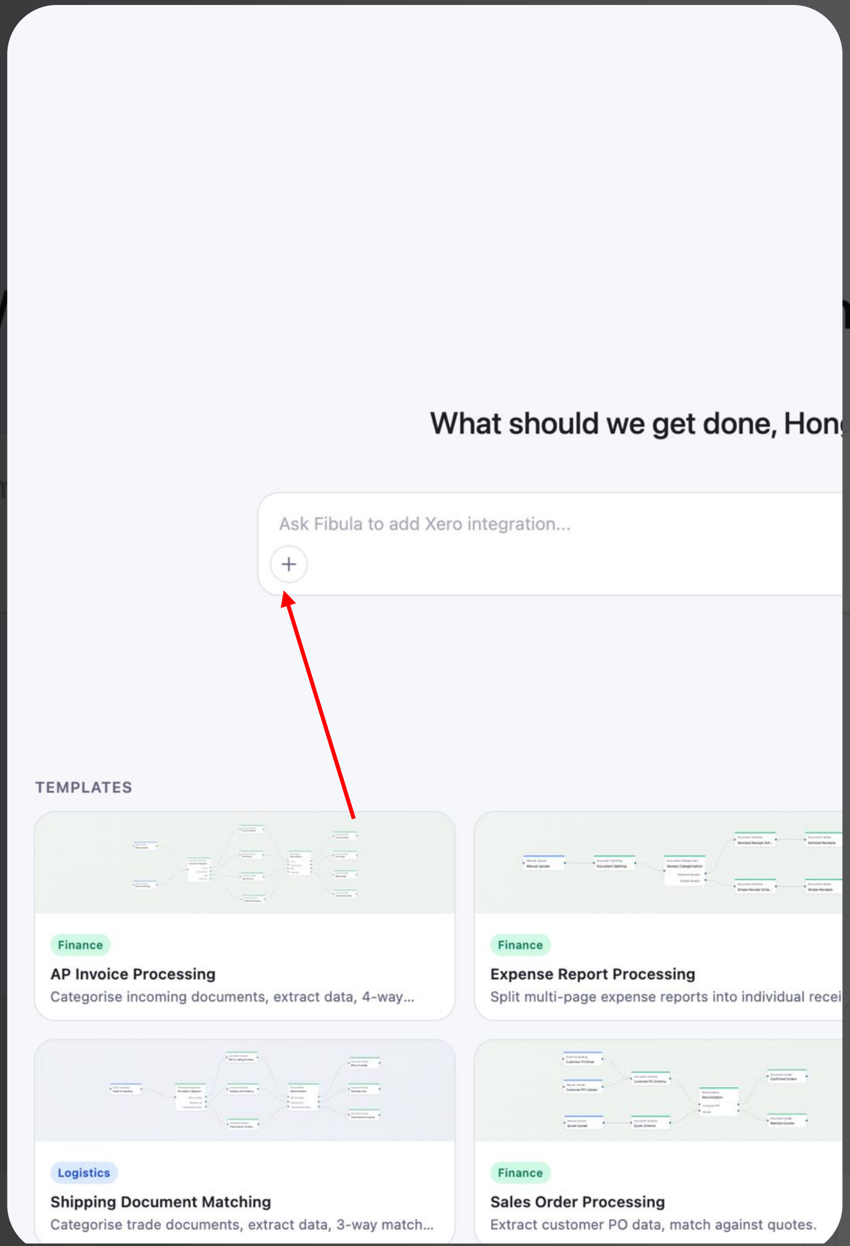
REPORTING

Findings are discussed with the team and Staple, then used to guide product improvements and follow-up testing in later sessions.

Problems will be tested in the new interview for validating the improvement

IMPROVE ENTRY POINT AND UPLOAD VISIBILITY

- Users are often struggle at the start of the workflow, starting point is not visible enough
- 42% task failure
- Users hesitate for a while



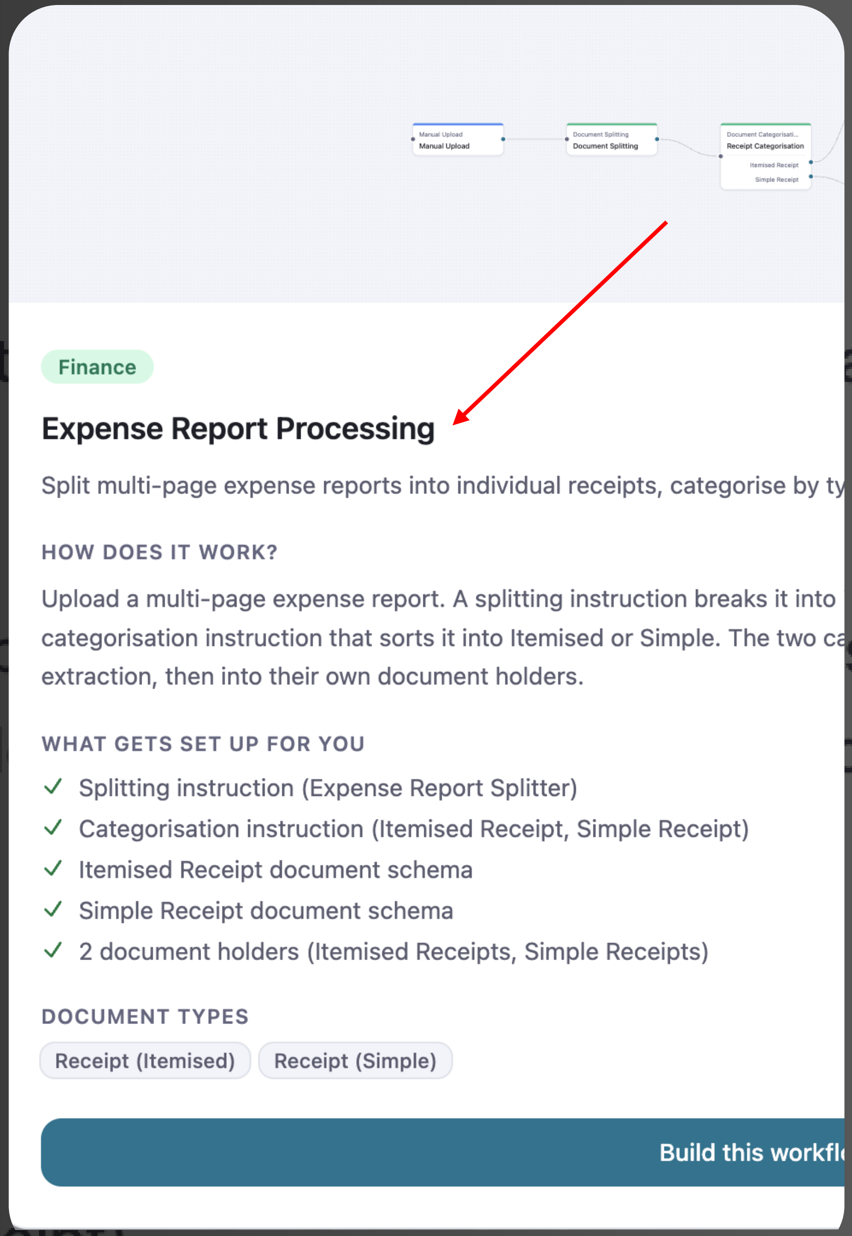
Recommendation:

- More visible upload entry point
- Side bar guidance
- Tooltip
- Clear start workflow and upload direction

Follow-up users moved into the workflow more smoothly without asking researcher where to start

REDUCE CHOICE OVERLOAD DURING SETUP

- User faced too many options and could not directly locate the template they wanted
- Too much information in template description
- 58% failure rate



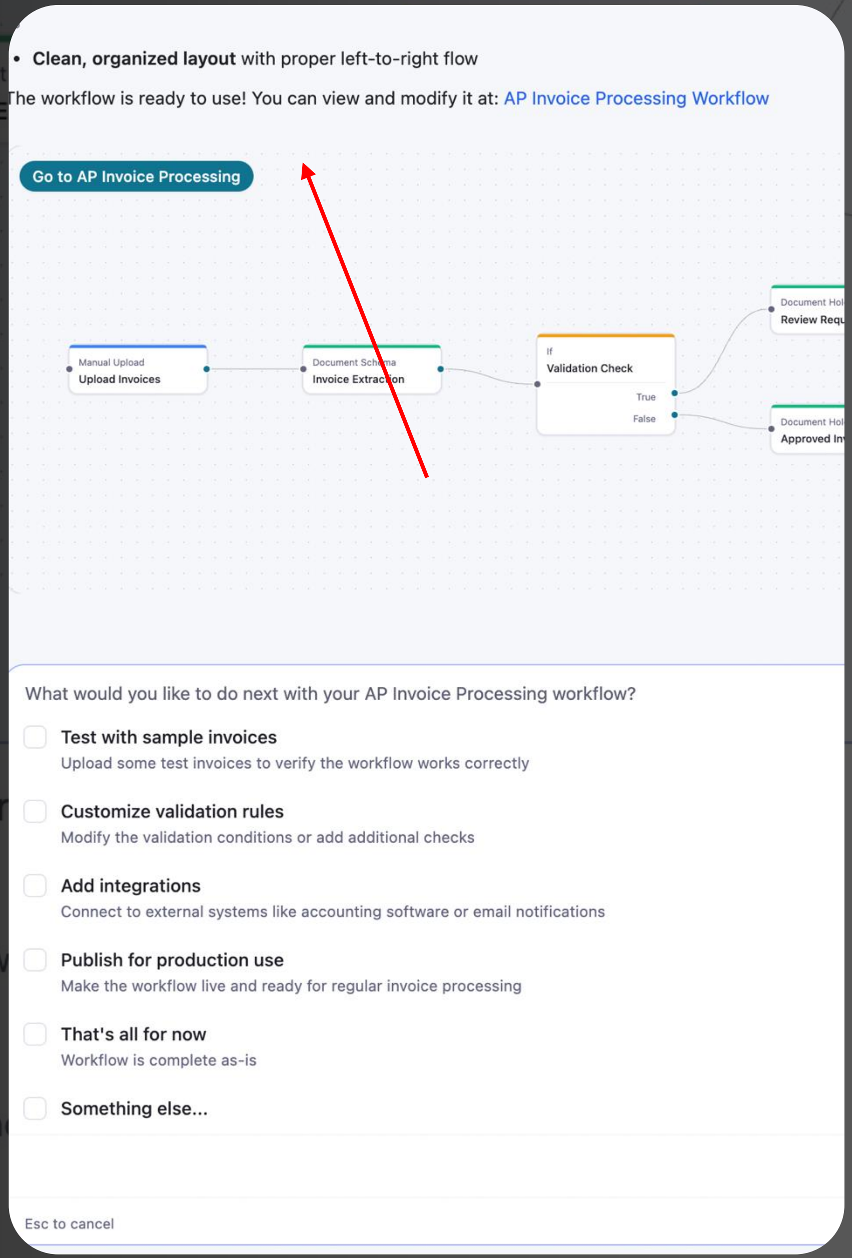
RECOMMENDATION:

- Reduce length of summary
- Template and keyword search
- Template grouping

Follow-up users indicated that the simplified setup flow made choices feel more manageable and reduced the tendency to select defaults only for speed.

IMPROVE NODE AND WORKFLOW LOGIC VISIBILITY

- User could not confidently understand how the generated nodes reflected their original business intent
- 73% failure
- Struggle to understand node



Recommendation:

- Clearer node visibility
- Tooltip
- Plain language explanation

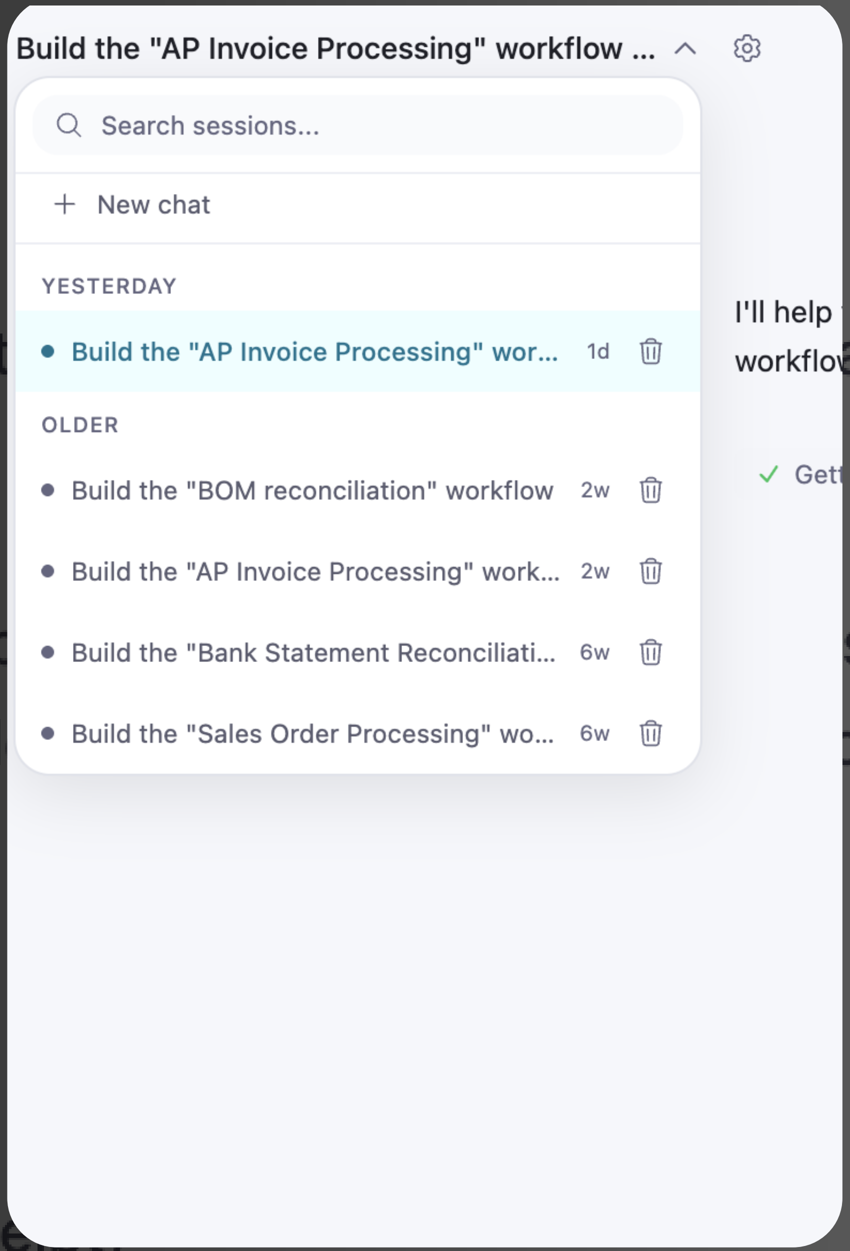
Follow-up users reported that clearer node status and document linkage made the generated workflow easier to understand no question similar to "so…what does this mean"

MAKE TESTING PAUSES RECOVERABLE

- User often run into failure, but they could not recover their work or does not know where to locate them
- 67% task failure rate

Recommendation:

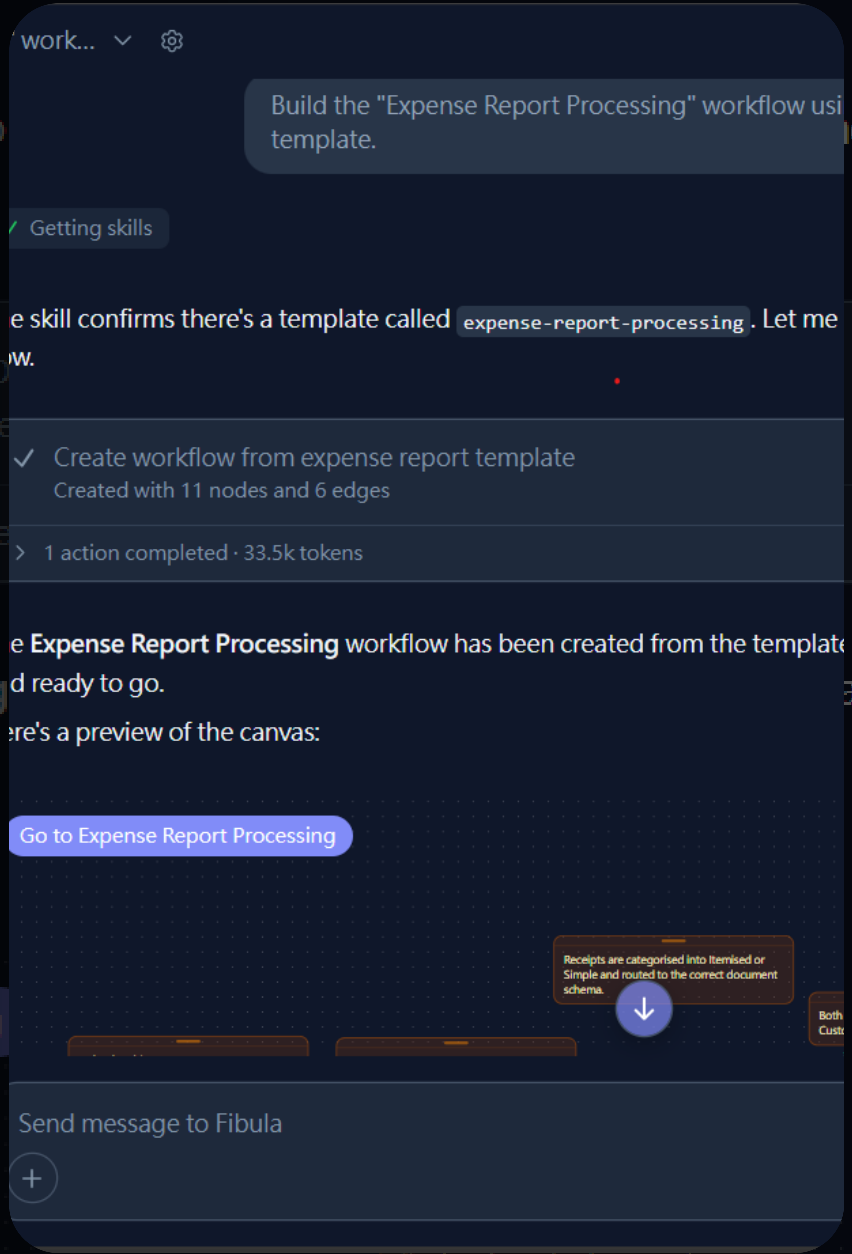
- Recoverable prompt
- History chat location
- Re-run option



Follow-up users treated testing pauses as recoverable steps because rerun, upload, and correction actions were more visible at the moment of failure, user did not stop when errors occurred

MAKE RECONCILIATION EVIDENCE ACTIONABLE

- Users had difficulty locating and acting on output evidence
- User had problem understanding the action done by system
- 51% task failure rate



Recommendation:

- Intermediate result/response
- Matched / unmatched summary
- More detailed guidance

Follow-up users found it easier to confirm reconciliation results, fewer participants questioned the system's respond to their task

Quantities Diagnostic Findings

Rate	Journey stage	Interaction risk	Severity
42%	Start	Hidden platform entry points	P2
58%	Template	Excessive, unstructutred choices	P1
73%	Configure	Ambigous node	P0
67%	Test	Hidden/unclear pipeline state	P0
51%	Output	Obscured exception	P2

Moderate friction, user completes task but hesitates, searches repeatedly, or misses key information

Major failure, user continues but chooses the wrong path or creates the wrong workflow

Critical blocker, user cannot continue without help

Core UX Finding

Fibula’s main UX issue was not total invisibility. The deeper problem was that users needed clearer explanations, better evidence surfacing, and smoother movement between chat, canvas, node details, and output states.

Spatial Desynchronization

Users moved between chatbot, canvas, testing view, and output panels, but these areas did not always feel connected.

Impact: Even when the workflow was visible, users could not always understand why the AI created certain nodes

Choice Faitgue

Long option lists, dense templates, and unfamiliar labels pushed users toward default or easiest choices

Impact: Users sometimes selected a workflow path that did not fully match their business intent

Operational Opacity

Users could see that a workflow paused, but not always the exact root cause, affected field, or required next action.

Impact: Recoverable pauses became no-help failures because users still needed manual support to diagnose the issue.

Challenges faced

While the project produced clear usability findings and product recommendations, the evaluation had several practical limitations due to the live, evolving nature of Fibula and the restricted access to company performance data.

Limited access to production data

Staple AI could not provide full production-level extraction accuracy, validation success rates, or large-scale processing logs. Therefore, the project focused on usability validation.

Product evolve during testing

Fibula was still under active development during the research period. Some interface elements, workflow functions, and recovery features changed between testing rounds.

Difference in user's workflow path

Participants entered Fibula through different routes, such as chatbot, builder, templates, or document interfaces. Some tested AP workflows, while others explored different automation tasks.

Hard to measure self-service

The no-help rule helped identify blockers, but real users may behave differently in workplace settings with training, documentation, or team support.

Application of design thinking

EMPHASIZE

Conducted task-based interviews and think-aloud observation to understand where users felt confused, overwhelmed, or blocked.

DEFINE

Reframed the problem from “Can AI extract documents?” to “Can users understand and act on AI-generated reconciliation workflows?”

IDEATE

Generated improvement ideas such as clearer workflow visibility, fewer setup choices, and natural-language recovery guidance.

PROTOTYPE

Mapped AP/BOM logic into Fibula workflows and proposed interface improvements

TEST

Used follow-up checks and later interviews to see whether revised interactions reduced friction.

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Applying AI in Smart Industry 4.0

Our project applied Industry 4.0 knowledge by using AI not only for document extraction but also as part of a stateful, human-in-the-loop workflow for industrial decision-making.

AI document understanding	Extracts structured information from unstructured PDFs, images, invoices, POs, GRNs, BOMs, and related documents.
LLM interaction	Allows users to describe reconciliation requirements in plain language rather than manually writing workflow logic.
Node-based workflow automation	Translates user intent into visible workflow steps, including ingestion, extraction, validation, exception routing, and review.
Human in the loop	Keeps humans involved when variance, uncertainty, or review decisions require judgment.
Explainnable workflow state	Shows documents, rules, variance reasons, and next actions so users can trust and audit the result.

Our achivements

Reconciliation logic formalized

We decomposed AP and BOM reconciliation into key documents, matching fields, validation checks, and review outcomes.

Fibula was evaluated with real users

Through task-based testing with participants, we identified where non-technical users struggled to create, test, and interpret workflows.

Key adoption barriers were diagnosed

The main issues were spatial desynchronization, choice fatigue, and operational opacity during configuration, testing, and output review.

Product improvements were proposed and validated

Recommendations included clearer workflow visibility, simplified setup choices, natural-language recovery guidance, and more actionable matched/unmatched outputs, and was tested through interviews.

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https://jungluchen.github.io/Innovation-Xlab/

References

[1] Staple AI. Document processing regulators can verify. https://www.staple.ai/
[2] Google Cloud. Staple AI processes tens of millions of documents with near-100% accuracy using Google AI. https://cloud.google.com/customers/staple-ai
[3] National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10
[4] Amershi, S., et al. (2019). Guidelines for Human-AI Interaction. CHI 2019. https://www.microsoft.com/en-us/research/publication/guidelines-for-human-ai-interaction/
[5] Object Management Group. (2014). Business Process Model and Notation (BPMN), Version 2.0.2. https://www.omg.org/spec/BPMN/2.0.2/
[6] Object Management Group. (2017). Unified Modeling Language (UML), Version 2.5.1. https://www.omg.org/spec/UML/2.5.1/
[7] Baviskar, D., Ahirrao, S., Potdar, V., & Kotecha, K. (2021). *Efficient automated processing of the unstructured documents using artificial intelligence: A systematic literature review and future directions*. IEEE Access, 9, 72894–72936.
[8] Saout, T., Lardeux, F., & Saubion, F. (2024). *An overview of data extraction from invoices*. IEEE Access, 12, 19872–19886.
[9] Rohaime, N. A., Abdul Razak, N. I., Thamrin, N. M., & Shyan, C. W. (2022). *Integrated invoicing solution: A robotic process automation with AI and OCR approach*. 2022 IEEE 20th Student Conference on Research and Development (SCOReD), 30–33. https://doi.org/10.1109/SCORED57082.2022.9973841
[10] Tater, T., Gantayat, N., Dechu, S., Jagirdar, H., Rawat, H., Guptha, M., Gupta, S., Strak, L., Kiran, S., & Narayanan, S. (2022). *AI driven accounts payable transformation*. Proceedings of the AAAI Conference on Artificial Intelligence, 36(11), 12405–12413.
[11] Dragomirescu, O.-A., Crăciun, P.-C., & Bologa, A. R. (2025). *Enhancing invoice processing automation through the integration of DevOps methodologies and machine learning*. Systems, 13(2), 87. https://doi.org/10.3390/systems13020087
[12] Baviskar, D., Ahirrao, S., & Kotecha, K. (2021). *Multi-layout unstructured invoice documents dataset: A dataset for template-free invoice processing and its evaluation using AI approaches*. IEEE Access, 9, 101494–101512. https://doi.org/10.1109/ACCESS.2021.3096739
[13] Katti, A. R., Reisswig, C., Guder, C., Brarda, S., Bickel, S., Höhne, J., & Faddoul, J. B. (2018). *Chargrid: Towards understanding 2D documents*. Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 4459–4469. https://doi.org/10.18653/v1/D18-1476
[14] Xu, Y., Li, M., Cui, L., Huang, S., Wei, F., & Zhou, M. (2020). *LayoutLM: Pre-training of text and layout for document image understanding*. Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 1192–1200. https://doi.org/10.1145/3394486.3403172
[15] Colakoglu, G., Solmaz, G., & Fürst, J. (2025). *Problem solved? Information extraction design space for layout-rich documents using large language models*. arXiv. https://arxiv.org/abs/2502.18179
[16] Yang, Q., Hu, Y., Cao, R., Li, H., & Luo, P. (2021). *Zero-shot key information extraction from mixed-style tables: Pre-training on Wikipedia*. Proceedings of the AAAI Conference on Artificial Intelligence, 35(2), 1451–1456. https://doi.org/10.1609/aaai.v35i2.16233
[17] Raja, S., Mondal, A., & Jawahar, C. V. (2022). *Visual understanding of complex table structures from document images*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2299–2308.
[18] Maurya, C. K., Gantayat, N., Dechu, S., & Horváth, T. (2020). *Online similarity learning with feedback for invoice line item matching*. arXiv. https://arxiv.org/abs/2001.00288
[19] Oral, B., & Eryiğit, G. (2022). *Fusion of visual representations for multimodal information extraction from unstructured transactional documents*. International Journal on Document Analysis and Recognition, 25, 187–205. https://doi.org/10.1007/s10032-022-00399-3
[20]Shneiderman, B. (2020). *Human-centered artificial intelligence: Reliable, safe & trustworthy*. International Journal of Human–Computer Interaction, 36(6), 495–504. https://doi.org/10.1080/10447318.2020.1741118
[21] n8n. (n.d.). *Nodes*. n8n Docs. https://docs.n8n.io/workflows/components/nodes/
[22] Langflow. (n.d.). *Langflow: A visual framework for building multi-agent and RAG applications*. GitHub. https://github.com/langflow-ai/langflow